3.3 COMPARISON OF VARIOUS FUELS

Commercially fuels are available in different grades like Unleaded Petrol, Speed Petrol, White Petrol, Diesel, Speed Diesel or Premium Diesel etc., Previously, Tetra Ethyl Lead (TEL) is mixed with petrol to increase the octane rating (for antiknocking). However, the lead emission emit from the vehicle is polluting the atmosphere and lead is also poisonous. Hence addition of TEL in petrol is banned and this petrol is called as Unleaded petrol. Various additives are added with fuel to enhance the properties. Such petrol will have high octane rating and called as Speed petrol or premium petrol. Similarly, additives are added with diesel to enhance the cetane rating and such diesel is called as Speed Diesel or Premium Diesel.

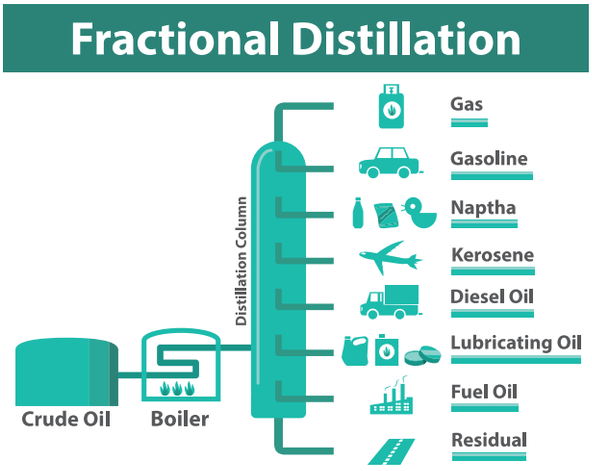


Figure 3.4 Fractional Distillation Process.

3.4 DISTILLATION CURVE

From the above curve, it is understood that the most volatile parts of the gasoline evaporate at lower temperature. This petrol vapor is mixes with air and makes the engine to start easy at cold condition. As the working temperature increases, the less volatile parts evaporate and mixes with air. Based on the distillation graph, the required additives during summer and winter season, can be added with fuel to ensure smooth operation of engine. Fig 3.4, 3.4(a) shows Distillation Curve and Fractional Distillation Process.

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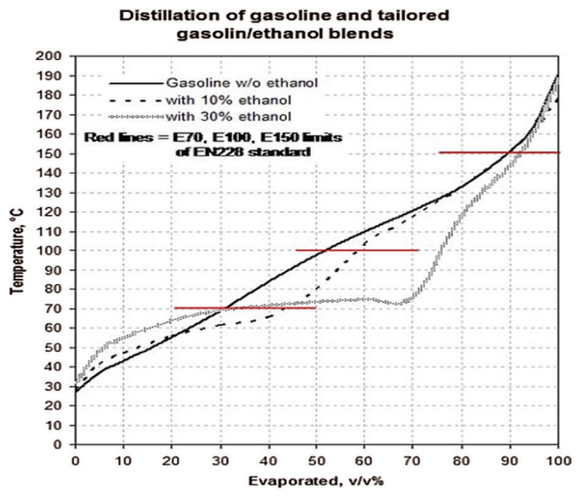


Figure 3.4(a) Distillation Curve-(% of Evaporation Vs Temperature °C)

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